

Money Plot Labs Technical Whitepaper

Optimal Transfer Cadence under Flat Fees

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Abstract

Contributions to a health savings account, brokerage sweep, or similar vehicle frequently accumulate as idle cash earning little or no return until they are transferred into investments. When each transfer carries a flat fee, the saver faces a tension: transferring often wastes money on fees, while transferring rarely forfeits compounding on cash left idle. We formalize the resulting trade-off, derive the final value of an arbitrary transfer schedule in closed form, and show that the optimal schedule can be found *exactly* at day resolution by a dynamic program in $O(k_{\max}T^2)$ time. We characterize the optimum as unimodal in the number of transfers, derive a practical “transfer threshold” rule of thumb from the optimal block sizes, and identify the low-fee boundary regime in which continuous transfer is optimal.

1 Introduction & Problem Framing

A saver contributes to an account whose cash balance earns a negligible return, but whose funds, once moved into investments, earn an expected annual return r . Each move from cash into investments incurs a flat fee f (for example, a \$25 partial-transfer fee). Two opposing forces define the problem. Cash left un-transferred suffers an **opportunity cost**: it sits at 0% while it could be compounding. But each transfer carries a **fee cost** that, paid early, also forgoes the compounding of the fee amount itself. The optimal policy balances these: enough transfers to keep capital working, but not so many that fees dominate. This paper derives that optimum exactly.

1.1 Parameter Definitions

We measure time in days over a fixed horizon of T days. The saver’s inflows are modeled as a smooth daily drip plus a set of discrete recurring lump sums:

- c — the daily contribution, $c = C_{\text{annual}}/365$, accruing as cash each day.
- $\mathcal{D} = \{(d_j, L_j)\}$ — lump sums of amount L_j deposited on day d_j , each recurring annually and expanded across the horizon.
- $\rho = r/365$ — the daily growth rate of invested funds.
- f — the flat fee charged per transfer.

The **cumulative nominal cash** contributed by day t (before any transfer) is

$$C(t) = ct + \sum_{(d_j, L_j) \in \mathcal{D}, d_j < t} L_j. \quad (1)$$

Because idle cash earns nothing, $C(t)$ is simply the running sum of contributions; all growth occurs only after a transfer.

1.2 The Growth Factor

A sum transferred into investments on day t compounds for the remaining $T - t$ days. We define the **growth factor**

$$g(t) = (1 + \rho)^{T-t}, \quad (2)$$

which is strictly decreasing in t : the earlier a dollar is transferred, the longer it compounds, so $g(t)$ is larger. At the horizon, $g(T) = 1$.

2 The Value of a Transfer Schedule

A transfer schedule is a set of k intermediate transfer days $0 = t_0 < t_1 < \dots < t_k < t_{k+1} = T$, where the final transfer at T is mandatory (the remaining cash is moved out at the end of the horizon). Transfer i captures all cash accrued since the previous transfer,

$$\Delta C_i = C(t_i) - C(t_{i-1}), \quad (3)$$

pays the fee, and the remainder compounds to the horizon at factor $g(t_i)$. Summing over all $k + 1$ transfers gives the final value:

$$V(t_1, \dots, t_k) = \sum_{i=1}^{k+1} (\Delta C_i - f) g(t_i), \quad g(t_{k+1}) = g(T) = 1. \quad (4)$$

(A block worth less than the fee contributes zero rather than a negative amount; at the optimum this clamp never binds, as no value-destroying transfer is ever selected.) The baseline of a *single* end-of-term transfer is the special case $k = 0$:

$$V_{\text{base}} = V() = C(T) - f. \quad (5)$$

3 Exact Optimization via Dynamic Programming

The naive search over all combinations of transfer days is $O(T^k)$ and quickly becomes infeasible; a coarse grid only approximates the optimum. However, the additive block structure of V admits an exact dynamic program. Let $\phi_j(t)$ denote the maximum value obtainable using exactly j transfers, the j -th occurring on day t , accounting for all cash from day 0:

$$\phi_1(t) = (C(t) - f) g(t), \quad (6)$$

$$\phi_j(t) = \max_{0 < p < t} \left[\phi_{j-1}(p) + (C(t) - C(p) - f) g(t) \right], \quad j \geq 2. \quad (7)$$

The factor $g(t)$ is shared by the entire block ending at t , which is what makes the recursion separable. For exactly k intermediate transfers, we append the mandatory final transfer at T :

$$V^*(k) = \max_{0 < p < T} \left[\phi_k(p) + (C(T) - C(p) - f) \right], \quad V^*(0) = C(T) - f. \quad (8)$$

The global optimum is $\max_k V^*(k)$, recovered with the corresponding schedule via backpointers. The procedure runs in $O(k_{\max} T^2)$ time at full day resolution — exact, and orders of magnitude faster than enumerating day-combinations. It reproduces the brute-force optimum to the cent on small cases while remaining tractable for long horizons.

4 The Sweet Spot: Unimodality in k

Adding one transfer splits an existing block, allowing the earlier sub-block to compound for additional days — a benefit that *diminishes* as blocks shrink — while incurring one more fee of effective cost $\approx f g(t)$. The marginal benefit decreases in k and the marginal cost is roughly constant, so $V^*(k)$ rises to a single peak k^* and then declines. Figure 1 shows this for a representative scenario (\$4,000/yr drip, a \$400 annual lump, $r = 4\%$, $f = \$25$, three-year horizon): value climbs steeply for the first few transfers, peaks at $k^* = 5$, and erodes thereafter as fees overtake the shrinking compounding gains.

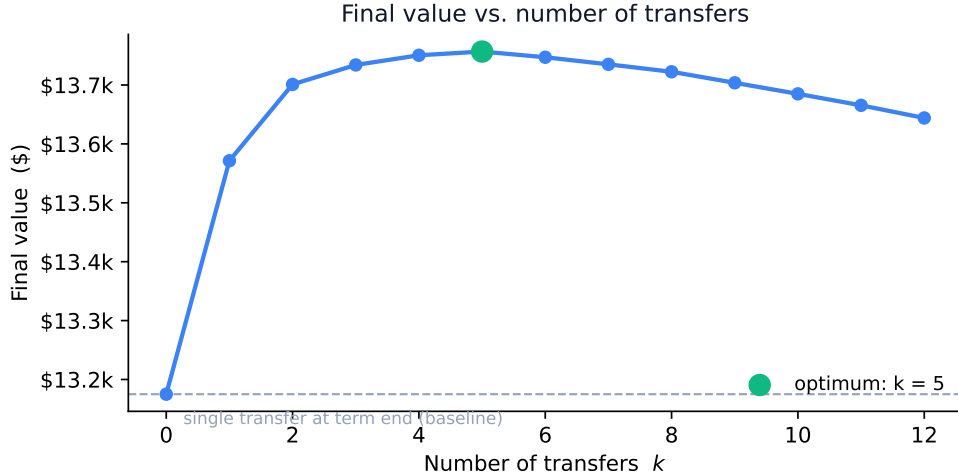


Figure 1: Final value as a function of the number of transfers. The optimum sits where the benefit of compounding capital sooner is exactly overtaken by the cost of additional fees. The dashed line marks the single-transfer baseline ($k = 0$).

5 The Idle-Cash Threshold

The optimal day-schedule is precise but awkward to follow in practice. A more usable policy emerges from the block sizes. The cash captured by transfer i , ΔC_i , equals the **idle-cash balance** just before that transfer fires. At the optimum these blocks are approximately equalized, which yields a simple operational rule:

$$\text{Transfer whenever idle cash reaches } \approx \overline{\Delta C}, \quad \overline{\Delta C} = \frac{1}{k} \sum_{i=1}^k \Delta C_i. \quad (9)$$

For the scenario above the optimal blocks are \$2,208, \$2,603, \$2,005, \$2,395, and \$1,995, giving a threshold near \$2,200. Figure 2 overlays the invested portfolio (rising) against the idle-cash sawtooth (accumulating, then dropping to zero at each transfer); the sawtooth peaks are exactly the block sizes, and a saver who simply transfers each time idle cash hits the threshold reproduces the optimum without tracking dates.

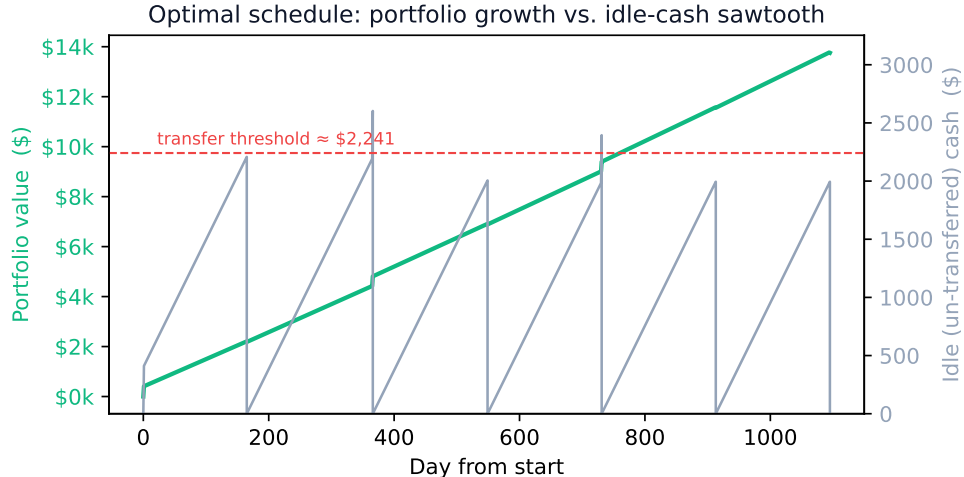


Figure 2: The optimal schedule, viewed as **invested value** (left axis) against the idle-cash sawtooth (right axis). Each tooth peaks at the transfer threshold (dashed **line**) and resets to zero when the transfer fires.

6 The Low-Fee Boundary Regime

The interior optimum k^* exists only when the fee is large enough to penalize frequency. As $f \rightarrow 0$, every additional transfer strictly increases the final value — moving any dollar into investments sooner compounds it longer at zero cost — so the optimum pushes toward the finest possible cadence. In the limit of a zero fee, the optimal policy is to transfer continuously: invest each contribution the instant it arrives. Practically, below a threshold fee the model’s recommended count grows without a meaningful interior peak, and the correct guidance is simply to transfer as often as is operationally convenient (e.g. every paycheck) rather than to chase a specific number.

7 Assumptions & Limitations

The model assumes a constant, deterministic expected return and therefore ignores market volatility and the path-dependence of real returns; it captures the *fee-versus-idle-cash* trade-off, not market timing. It omits taxes, and it does not encode account-specific rules such as minimum invested balances or transfer minimums. Contributions are idealized as a smooth daily drip plus discrete recurring lumps, and the fee is treated as a flat per-transfer charge. The output is best read as guidance on *cadence* — how often, and at what idle-cash level, to transfer — rather than as a prescription of exact calendar dates. This document is an analytical model, not tax or investment advice.

8 Conclusion

Treating fee-bearing capital transfers as a partitioning problem turns a vague “how often should I move my cash?” into an exactly solvable optimization. The final value of any schedule decomposes into independent compounding blocks, a dynamic program locates the global optimum at day resolution in polynomial time, and the optimum is unimodal in the number of transfers. Most usefully for practice, the optimal block sizes collapse into a single threshold rule: transfer whenever idle

cash reaches a characteristic amount. The accompanying interactive tool computes this optimum, schedule, and threshold for arbitrary contribution patterns and fees.